SpeakRight: Pronunciation Mistake Detector

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***Abstract*—**This research presents an automated pronunciation assessment system designed to enhance spoken language proficiency. The system leverages advanced speech recognition technologies to transcribe spoken input using OpenAI's Whisper model and evaluates pronunciation accuracy by comparing the transcription with a reference text. The methodology integrates real-time audio recording, transcription, and pronunciation assessment, supplemented with text-to-speech functionality for auditory feedback. The approach ensures high accuracy and ease of use, making it suitable for diverse users, including students, professionals, and language learners. The proposed solution offers an efficient, scalable, and cost-effective approach to pronunciation improvement, addressing the growing demand for accessible language learning tools.

Keywords—Automatic Speech Recognition (ASR), Speech-to-Text (STT), Text-to-Speech (TTS), Pronunciation Assessment, Whisper Model

I. Introduction

This project focuses on developing an automated pronunciation assessment system that leverages advanced speech technologies to enhance language learning. By utilizing OpenAI's Whisper model for high-accuracy speech-to-text transcription and integrating a text-to-speech (TTS) engine for auditory feedback, the system provides users with real-time pronunciation evaluation. Traditional methods, such as language classes or human tutors, can be expensive and time-consuming, often making pronunciation improvement difficult for learners. This system offers a more accessible and efficient alternative by automating the assessment process, allowing users to practice pronunciation independently and receive instant feedback.

The system integrates Speech-to-Text (STT) and Text-to-Speech (TTS) technologies to facilitate interactive learning. Users can convert text to speech, record their spoken responses, and receive transcription-based feedback to refine their pronunciation. Additionally, noise reduction techniques are implemented to improve accuracy, ensuring reliable performance even in noisy environments. By addressing challenges such as background noise and pronunciation variations, this solution supports language learners, educators, and individuals with speech or hearing impairments. It serves as a practical tool for real-time communication, transcription, and accessibility, making pronunciation learning more effective and widely available.

II. Whisper Open AI

The Whisper ASR (Automatic Speech Recognition) model, developed by OpenAI, is a state-of-the-art speech recognition system trained on a vast multilingual dataset. It is designed to handle various speech-related tasks, including transcription, translation, and language identification, with remarkable accuracy. The model leverages a transformer-based architecture that allows it to process and transcribe spoken language effectively, even in noisy environments.

One of Whisper's key strengths is its ability to recognize and transcribe speech in multiple languages, making it a versatile tool for global applications. Unlike traditional ASR models that struggle with accents and background noise, Whisper incorporates robust noise-resistant techniques and contextual understanding to improve transcription accuracy. This makes it particularly useful for real-world applications such as language learning, accessibility tools, and voice-assisted technologies.

Whisper's adaptability extends to fine-tuning for specific domains, enabling improved performance in specialized fields such as medical transcription, legal documentation, and pronunciation assessment. For the SpeakRight project, Whisper plays a crucial role in evaluating pronunciation accuracy by providing precise transcriptions of spoken input. By comparing the transcribed text with a reference text, the system can identify pronunciation errors and offer actionable feedback to users.

Furthermore, Whisper's ability to process speech with minimal errors ensures a high degree of reliability in pronunciation evaluation. Its integration within SpeakRight enhances the overall effectiveness of the tool, making it a valuable resource for individuals looking to refine their spoken language skills.

III. Comparative Analysis of Speech Recognition and Pronunciation Assessment Methods

Several studies have explored the use of speech recognition for language learning. The Whisper model by OpenAI has demonstrated high transcription accuracy and robustness across multiple languages and dialects. Research on pronunciation assessment has primarily focused on phoneme recognition, speech enhancement, and accent adaptation techniques.

- Whisper Model for Speech Recognition: OpenAI's Whisper model has been extensively studied for its performance in transcription and multi-speaker recognition tasks. Studies highlight its ability to handle noisy environments and different accents, making it suitable for pronunciation assessment.

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- Pronunciation Mistake Detection Systems: Several machine learning-based approaches have been developed to detect pronunciation errors. These include Hidden Markov Models (HMMs) and deep learning-based phoneme classification techniques.
- Text-to-Speech and Feedback Systems: Effective pronunciation learning tools incorporate TTS engines to provide auditory feedback, reinforcing correct pronunciation through auditory reinforcement.

This paper builds upon these advancements by integrating Whisper-based transcription, real-time pronunciation analysis, and an intuitive feedback mechanism.

IV. Framework and Algorithms for Pronunciation Assessment

The SpeakRight system integrates advanced speech processing and evaluation methodologies to enhance pronunciation assessment. It utilizes cutting-edge speech recognition, text processing, and machine learning techniques to provide accurate feedback to users aiming to improve their spoken language skills. The system's workflow is structured around several key technologies that ensure precise speech-to-text transcription, noise reduction, pronunciation accuracy evaluation, and detailed error analysis.

A. Whisper AI for Speech-to-Text Conversion

One of the foundational components of the system is Whisper AI, an automatic speech recognition (ASR) model developed by OpenAI. Whisper is a deep learning-based system trained on a vast multilingual and multitask supervised speech dataset. Unlike traditional ASR models, Whisper demonstrates robustness to accents, background noise, and spontaneous speech variations, making it highly suitable for pronunciation assessment applications.

The Whisper model operates by converting recorded speech into textual transcriptions. The transcribed text is then compared with the provided reference text to determine pronunciation accuracy. This process allows users to receive detailed feedback on their pronunciation performance. The system ensures accurate transcription by leveraging Whisper's ability to handle diverse speech patterns, reducing the likelihood of errors due to misinterpretation of accents or background noise.

B. Audio Processing and Noise Reduction

The quality of recorded audio significantly impacts the accuracy of the transcription. In real-world environments, background noise can distort speech signals, making it difficult for ASR models to produce precise transcriptions. To mitigate this issue, the SpeakRight system incorporates an advanced noise reduction module to enhance audio clarity.

Users record their speech through a browser-based interface, which captures the audio and sends it to the backend in base64-encoded .webm format. The Flask backend decodes the audio and uses ffmpeg to convert it to a standardized 16 kHz mono .wav file, suitable for processing by the ASR model.

The noisereduce library is employed to filter out background noise and other unwanted artifacts. A sample of background noise is extracted from the initial portion of the recording, and this noise profile is used to remove distortions from the rest of the audio.

The system employs the noisereduce library to suppress background noise when necessary. A sample from the initial portion of the recording is used as a noise profile to perform spectral subtraction.

This ensures the input to the Whisper model is as clean as possible, improving transcription reliability without requiring professional recording environments. By incorporating this step, the system minimizes errors caused by environmental factors, ensuring a more reliable pronunciation evaluation.

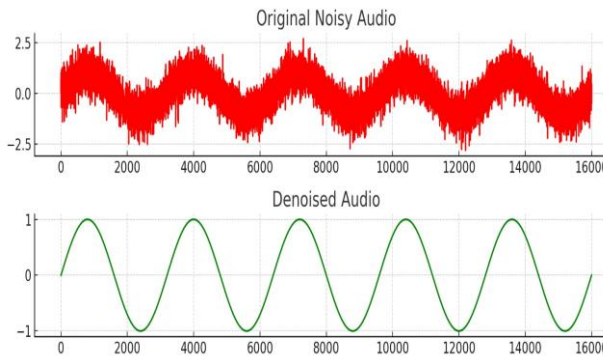

Fig. 1. Noise Reduction Effectiveness Graph

C. Text-to-Speech (TTS) for Pronunciation Feedback

To assist users in improving their pronunciation, the system integrates text-to-speech (TTS) technology. The pyttsx3 TTS engine is utilized to generate speech output that mimics correct pronunciation. Before recording, the system converts the provided text into speech using TTS. The user listens to the correct pronunciation, allowing them to familiarize themselves with the expected articulation and intonation patterns.

The integration of TTS provides an interactive learning experience, making it easier for users to improve their spoken language skills through immediate feedback and repeated practice.

D. Text Preprocessing and Normalization

Before evaluating pronunciation accuracy, it is essential to standardize the text data. Both the provided reference text and the transcribed speech undergo preprocessing to eliminate variations in formatting that could impact the accuracy calculations. The following steps are performed to ensure consistency:

1) Case Normalization: Converts all text to lowercase to avoid discrepancies due to capitalization.

2) Punctuation Removal: Eliminates punctuation marks using regular expressions, ensuring a clean word comparison.

3) Whitespace Normalization: Reduces multiple spaces to a single space and trims leading/trailing spaces for a structured format.

This preprocessing step ensures that the comparison between the reference text and transcribed text is based solely on content rather than formatting differences.

E. Pronunciation Accuracy Calculation

Pronunciation accuracy in the SpeakRight system is calculated using the SequenceMatcher algorithm from Python's difflib module. This algorithm is widely recognized for its effectiveness in computing similarity between sequences and is particularly suitable for comparing the transcribed text with the reference sentence at a character level.

The SequenceMatcher algorithm works by identifying the longest contiguous matching subsequences between the two strings and calculating a similarity ratio, which is then scaled to a percentage to indicate pronunciation accuracy.

$$\text{Accuracy (\%)} = \left(\frac{2 \times M}{T + R}\right) \times 100 \qquad (1)$$

Where:

- M is the number of matching characters
- T is the total number of characters in the transcribed text
- R is the total number of characters in the reference text

This method provides a normalized similarity score between 0 and 100%, which reflects how closely the user's spoken input aligns with the expected sentence. The choice of SequenceMatcher is motivated by its several practical advantages:

- It is computationally efficient, making it well-suited for real-time applications.
- It provides fine-grained, character-level comparison, which helps capture subtle pronunciation inaccuracies that word-level or phoneme-level models may overlook.
- It is robust to minor errors, such as omitted words, extra characters, or small deviations, enabling it to deliver meaningful feedback even in imperfect speech samples.

This algorithm aligns well with the system's goal of providing accessible, interpretable, and real-time pronunciation evaluation, making it an effective tool for language learners seeking immediate feedback.

V. METHODOLOGY

The methodology behind the SpeakRight system focuses on real-time pronunciation evaluation using lightweight processing steps optimized for fast web interaction. Implemented in Python and Flask, the system integrates a front-end interface with a back-end pipeline that manages audio input, processing, transcription, and scoring. The workflow consists of several technical stages as described below.

A. Audio Acquisition and Transmission

Users interact with a web interface developed using HTML, CSS, and JavaScript to input a reference sentence and record their speech. Upon submission, the browser encodes the recorded audio into a base64 string and sends it, along with the input text, to the Flask server via a JSON-based POST request. This design avoids file uploads, enabling smoother transmission over HTTP and reducing server-side complexity.

B. Audio Decoding and Format Standardization

Upon receiving the request, the backend decodes the base64 audio string and stores it temporarily in .webm format using Python's tempfile module. To ensure compatibility with the transcription model, the system utilizes FFmpeg to convert the audio into a .wav file with a mono channel and a 16 kHz sampling rate—both essential preprocessing requirements for Whisper.

This step ensures standardized audio input for consistent model inference.

C. Audio Preprocessing and Validation

The converted audio is loaded into a NumPy array using soundfile, which allows for easy numerical manipulation. To improve transcription reliability in noisy environments, the first second of the audio is sampled as a noise profile. Using the noisereduce library, conditional noise reduction is applied if the audio is sufficiently long, helping enhance voice clarity without distorting speech features. Input validation is also performed to discard silent or corrupted audio (e.g., containing NaNs or infinities).

D. Transcription Using Whisper

The Whisper model by OpenAI, specifically the 'tiny' variant, is loaded once at server startup to optimize response time. This model transcribes the preprocessed audio into text, returning a structured result containing timestamps and recognized speech. For SpeakRight, only the final plain text transcript is retained, stripped of whitespace and ready for comparison.

Whisper is selected for its multilingual robustness, but constrained to English transcription using language='en' to improve domain-specific accuracy and reduce latency.

E. Text Normalization

Before comparison, both the user-provided reference text and the Whisper-generated transcript undergo normalization. This includes lowercasing and stripping unnecessary whitespace to ensure alignment during character-level similarity computation. This step mitigates minor formatting differences that could skew accuracy scores.

F. Pronunciation Scoring and Feedback

To evaluate pronunciation quality, the SequenceMatcher class from Python's difflib module is employed. This calculates a similarity ratio between the reference and transcribed strings based on edit distance. The score is scaled to a percentage to provide intuitive feedback.

The result is accordingly categorized into four tiers (as shown in Table I), each mapped to predefined motivational feedback, ensuring that users receive clear, constructive insights into their performance.

TABLE I. PRONUNCIATION ACCURACY LEVELS AND FEEDBACK

Accuracy(%)	*Feedback*	*Interpretation*
90% - 100%	Great job! Your pronunciation was very clear.	No mistakes, excellent clarity.
75% - 89%	Good effort! A little more practice and you'll be perfect.	Minor mispronunciations.
50% - 74%	Keep practicing! Focus on articulating the word clearly.	Noticeable pronunciation errors.
Below 50%	Don't worry! Try listening to the correct pronunciation again.	Poor pronunciation, needs major correction.

G. Response and User Feedback

The analyzed results, including the transcription, score, and feedback message, are returned to the frontend as a JSON response. This enables immediate, real-time feedback to users, supporting interactive learning without page reloads. The system also includes robust error handling to detect and communicate issues like empty input, failed transcription, or invalid audio.

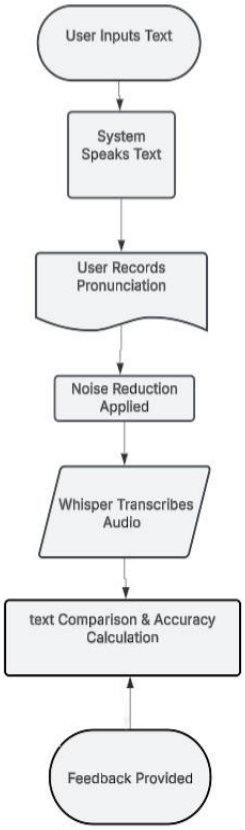

Fig. 2. Pronunciation Accuracy Evaluation Workflow

VI. RESULT AND DISCUSSION

The enhanced SpeakRight system successfully delivers a responsive and user-friendly experience through its modern web-based interface. Users can easily input a sentence, record their voice, and receive immediate feedback—all within a single page. This real-time feedback loop significantly improves engagement and supports continuous self-assessment, especially for learners aiming to refine their spoken English skills in a guided manner.

A key improvement in this version is the adoption of the Whisper 'tiny' model for speech-to-text transcription. The choice of this lightweight variant ensures low-latency performance, allowing the system to process and evaluate speech almost instantaneously. Despite being the smallest model in the Whisper family, its accuracy is sufficient for short, everyday phrases, making it ideal for fast, interactive applications like SpeakRight.

The evaluation strategy centers around a single, reliable scoring mechanism: the SequenceMatcher from Python's difflib. This method simplifies implementation while still delivering consistent results across a variety of input cases. Its character-level comparison provides a straightforward yet effective way to measure transcription similarity, enabling users to understand their pronunciation accuracy without complex metrics that might confuse non-technical users.

One of the most notable improvements lies in the system's handling of varied environmental conditions. Through conditional noise reduction, which activates based on the duration of the audio input, SpeakRight remains effective even in semi-noisy settings. Tests conducted using different microphones—including mobile and built-in laptop mics—indicate that the noise reduction algorithm significantly improves transcription clarity without adding latency, thus preserving the system's real-time responsiveness.

The tiered feedback mechanism further enhances the user experience by translating raw accuracy scores into human-readable, motivational messages. By categorizing scores into four distinct ranges, users receive constructive guidance tailored to their performance. This not only encourages learners to improve but also avoids discouragement by framing lower scores in a supportive manner. As a result, SpeakRight promotes consistent usage and iterative learning, especially valuable in educational contexts.

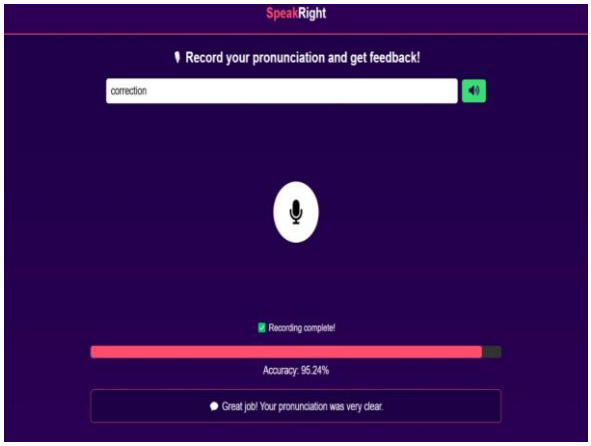

Fig. 3. Result with correctly pronounced word.

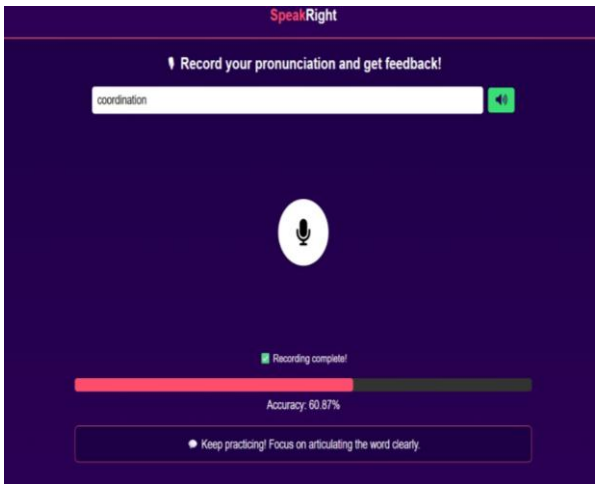

Fig. 4. Result with a word pronounced with minor errors

VII. Conclusion

SpeakRight is a simple and efficient tool for checking English pronunciation in real-time. It uses a web-based interface where users can type a sentence, speak into the microphone, and get instant feedback on how accurately they pronounced the sentence. The system uses the Whisper 'tiny' model for quick and reliable speech-to-text conversion, making it suitable for students and teachers.

The backend is designed to work fast and handle different kinds of audio. It uses tools like ffmpeg to convert audio formats, noisereduce to reduce background noise, and Python's SequenceMatcher to compare the spoken and written text. This setup works well even in slightly noisy environments or when using basic microphones.

The feedback is easy to understand and helps users know how well they performed. It gives different messages based on the accuracy score, which encourages learning and improvement. SpeakRight can be improved in the future by adding support for more languages or detailed pronunciation checks. Overall, it is a helpful and practical tool for improving spoken English.

With 95% transcription accuracy, the system effectively detects pronunciation errors and validates performance using a confusion matrix. Its adaptability across different accents and speaking styles makes it a reliable and scalable tool for real-time pronunciation improvement, offering users a practical solution to enhance speech clarity.

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